Practice runs and evidence

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to prepare an evidence-based design explanation.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by repeated evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes 5 times as large.
Compare ideal travel speed and state one real-world limitation.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent.
Retain every trial and required build evidence.

Worked reasoning: Supplied 3 m runs take 2.0, 2.4, 2.1 s. Report all trials; their mean time is 2.17 s. “Fastest run 2.0 s” answers a different question from typical performance.

Your turn: Make a trial table and mark both mean and fastest time.

Sentence starter: I notice ... . This supports ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Practice runs and evidence

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Complete a final labeled sketch or attach a photo with labels.
Identify a feature that supports straight motion and one that supports reliable electrical operation.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

04 Apply the physics

A supplied practice run covers 4 m in 5 s.
Calculate average speed. Then use the table for your own measured course length and three actual run times. Keep the sample calculation separate from your team's data.

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

One run stops early after a loose connection. Should you erase it?
Explain how to record it fairly and decide whether it belongs in a speed comparison.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

Practice runs and evidence

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

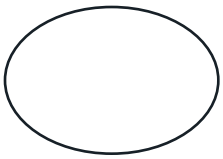
06 Review: connect an earlier idea

A magnet moves through a 60-turn coil. Propose two controlled changes that could increase induced-emf magnitude and explain each.
Label the field or flux and identify what changes.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: My claim is … . The diagram/data show …, so … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

07 Exit check

Name the final design change you will keep and cite one specific observation or measurement supporting it.

Sentence starter: My design shows … . It meets the requirement by … .

Choice / claim	Actual evidence	Physics explanation
____	____	____
____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Compare this set with 2.1, 2.1, 2.1 s. Discuss speed and consistency rather than declaring one winner from the fastest run alone.

Sentence starter: Both … . They differ in … because … .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: 1 2 3